WORKED ANSWERS

Inequalities

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Remember: *> greater than, < less than, ≥ greater than or equal to, ≤ less than or equal to. On a number line, a* ***closed dot*** *● means the number is included and an* ***open dot*** *○ means it is not included. An inequality is solved like an equation: the same number can be added to or subtracted from both sides, and both sides can be multiplied or divided by the same* ***positive*** *number. If both sides are multiplied or divided by a* ***negative*** *number, the inequality sign must be* ***reversed****.*

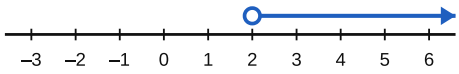
Question 1

(i) $x > 5$ (ii) $y \leq -2$ (iii) $a \geq 18$ (iv) $m \leq 20$ (v) $20 \leq t \leq 30$

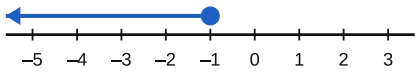
Question 2

(i) True (ii) False (iii) True (iv) True (v) False ($-6 < -5$)

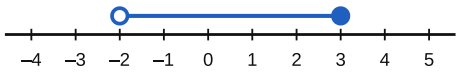
Question 3



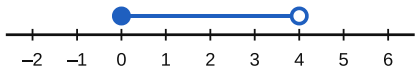
(i) $x > 2$



(ii) $x \leq -1$



(iii) $-2 < x \leq 3$



(iv) $0 \leq x < 4$

Question 4

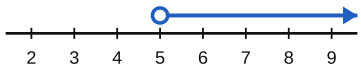
(i) $x \geq 1$ (ii) $x < 3$ (iii) $-3 \leq x < 2$ (iv) $-1 < x < 4$

Question 5

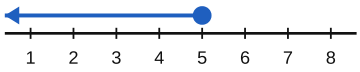
(i) $-1, 0, 1, 2, 3$ (ii) $-4, -3, -2, -1$ (iii) $1, 2, 3, 4, 5$

Question 6

(i) $x > 5$ (ii) $x \leq 5$ (iii) $x \geq 4$ (iv) $x < 6$ (v) $2x > 6 \rightarrow x > 3$ (vi) $3x \leq 12 \rightarrow x \leq 4$



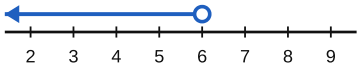
(i)



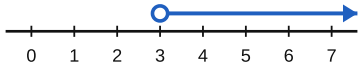
(ii)



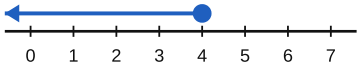
(iii)



(iv)



(v)



(vi)

Question 7

(i) Divide both sides by -2 and reverse the sign: $\mathbf{x > -3}$

(ii) $-x \geq -3 \rightarrow$ multiply by -1 and reverse the sign: $\mathbf{x \leq 3}$

Question 8

(i) $3x - 6 < x + 4 \rightarrow 2x < 10 \rightarrow \mathbf{x < 5}$; greatest integer = **4**

(ii) $x + 1 \geq 6 \rightarrow \mathbf{x \geq 5}$; smallest integer = **5**

Question 9

(i) $45n \leq 500 \rightarrow n \leq 11.1\ldots \rightarrow$ maximum **11 pens**

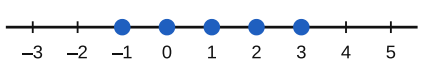
(ii) $70 + 40n \leq 600 \rightarrow 40n \leq 530 \rightarrow n \leq 13.25 \rightarrow$ maximum **13 boxes**

(iii) $n + (n + 1) + (n + 2) < 40 \rightarrow 3n + 3 < 40 \rightarrow n < 12.33\ldots \rightarrow$ largest n = **12**; integers **12, 13, 14** (sum = 39)

Question 10

(i) $-2 < x \leq 3 \rightarrow$ **−1, 0, 1, 2, 3**

(ii) **2, 4, 6**



(i)

Question 11

(i) $2(8 + x) < 30 \rightarrow 16 + 2x < 30 \rightarrow \mathbf{x < 7}$ (and $x > 0$, since a breadth must be positive)

(ii) **1, 2, 3, 4, 5, 6**